

Experiment - 04 (HomeTask)

Branch: BE-CSE(AIML) **Subject:** Advanced Database Management System

Question 1: Product Line Revenue

Source: DataLemur — Product Line Revenue — <https://datalemur.com/questions/revenue-by-product-line>

As a business analyst on the revenue forecasting team at NVIDIA, you are given a table of NVIDIA transactions in 2021. Write a query to summarize the total sales revenue for each product line. The product line with the highest revenue should be at the top of the results. Assume there will be at least one sale of each product line.

products Table:

Column Name	Type
product_id	integer
product_name	varchar
product_line	varchar

products Example Input:

product_id	product_name	product_line
1	Quadro RTX 8000	GPU
2	Quadro RTX 6000	GPU
3	GeForce RTX 3060	GPU
4	BlueField-3	DPU

transactions Table:

Column Name	Type
transaction_id	integer
product_id	integer
amount	integer

transactions Example Input:

transaction_id	product_id	amount
101	1	5000
102	2	4200
103	3	9000
104	4	7000

Example Output:

product_line	total_revenue
GPU	18200
DPU	7000

Solution:

```
SELECT
    P.PRODUCT_LINE,
    SUM(T.AMOUNT) AS TOTAL_REVENUE
FROM TRANSACTIONS AS T
JOIN PRODUCTS AS P
    ON T.PRODUCT_ID = P.PRODUCT_ID
GROUP BY P.PRODUCT_LINE
```

ORDER BY TOTAL_REVENUE DESC;

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Question 2: International Call Percentage

Source: DataLemur — International Call Percentage — <https://datalemur.com/questions/international-call-percentage>

A phone call is considered an international call when the person calling is in a different country than the person receiving the call. Write a query to find what percentage of phone calls are international. Round the result to 1 decimal place. Assume the caller_id column in the phone_info table refers to both the caller and the receiver.

phone_calls Table:

Column Name	Type
caller_id	integer
receiver_id	integer
call_time	timestamp

phone_calls Example Input:

caller_id	receiver_id	call_time
1	2	2022-07-04 10:13:49
1	5	2022-08-21 23:54:56
5	1	2022-05-13 17:24:06
5	6	2022-03-18 12:11:49

phone_info Table:

Column Name	Type
caller_id	integer
country_id	varchar
network	varchar
phone_number	varchar

phone_info Example Input:

caller_id	country_id	network	phone_number
1	US	Verizon	+1-212-897-1964
2	US	Verizon	+1-703-346-9529
3	US	Verizon	+1-650-828-4774
4	US	Verizon	+1-415-224-6663
5	IN	Vodafone	+91 7503-907302
6	IN	Vodafone	+91 2287-664895

Example Output:

international_calls_pct
50.0

Solution:

```
SELECT  
    ROUND(  
        100.0 * SUM(CASE WHEN C1.COUNTRY_ID != C2.COUNTRY_ID THEN 1 ELSE 0  
    END)
```

```
        / COUNT(*), 1
    ) AS INTERNATIONAL_CALLS_PCT
FROM PHONE_CALLS AS PC
JOIN PHONE_INFO AS C1
    ON PC.CALLER_ID = C1.CALLER_ID
JOIN PHONE_INFO AS C2
    ON PC.RECEIVER_ID = C2.CALLER_ID;
```

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